

# Rotation Matrix in 2D — Derivation via Trig Identities

## 1 Goal

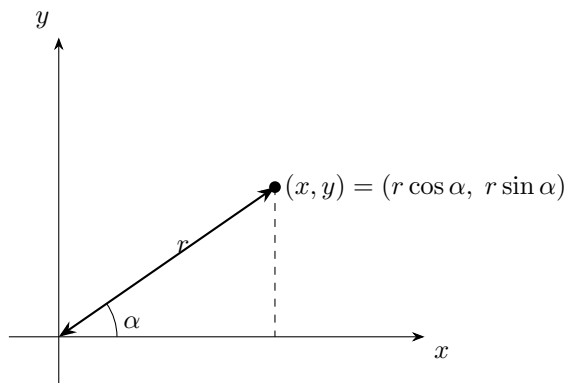
We want a matrix  $R(\theta)$  such that multiplying it by a vector  $(x, y)$  gives the vector  $(x', y')$  obtained by rotating  $(x, y)$  counterclockwise by an angle  $\theta$  about the origin. We derive it from *polar coordinates* and the *angle-addition (sum) identities* for sine and cosine.

## 2 Idea: think of the point in polar coordinates

Any point  $(x, y) \neq 0$  has polar form

$$x = r \cos \alpha, \quad y = r \sin \alpha,$$

where  $r = \sqrt{x^2 + y^2}$  is the distance from the origin and  $\alpha$  is the angle the ray to  $(x, y)$  makes with the positive  $x$ -axis.



Rotating the point by  $\theta$  means the new point is at angle  $\alpha + \theta$  from the positive  $x$ -axis, with the *same* radius  $r$  (rotation preserves length). So

$$x' = r \cos(\alpha + \theta), \quad y' = r \sin(\alpha + \theta).$$

## 3 Apply the angle-addition identities

The sum identities for sine and cosine are

$$\cos(\alpha + \theta) = \cos \alpha \cos \theta - \sin \alpha \sin \theta, \quad \sin(\alpha + \theta) = \sin \alpha \cos \theta + \cos \alpha \sin \theta.$$

Substitute these into  $x'$  and  $y'$ :

$$x' = r \left( \underbrace{\cos \alpha}_{x/r} \cos \theta - \underbrace{\sin \alpha}_{y/r} \sin \theta \right) = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta = x \cos \theta - y \sin \theta,$$

$$y' = r \left( \underbrace{\sin \alpha}_{y/r} \cos \theta + \underbrace{\cos \alpha}_{x/r} \sin \theta \right) = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta = x \sin \theta + y \cos \theta.$$

We used  $x = r \cos \alpha$  and  $y = r \sin \alpha$  again to replace  $r \cos \alpha \rightarrow x$  and  $r \sin \alpha \rightarrow y$ .

## 4 Extract the matrix

Collect the two equations as a single matrix-vector product:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{R(\theta)} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Therefore the **2D rotation matrix** (counterclockwise by  $\theta$ ) is

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

The minus sign on the top right comes from  $\cos(\alpha + \theta)$  having the minus; the off-diagonal sines keep the same sign from  $\sin(\alpha + \theta)$ .

## 5 The geometry: why the columns are unit vectors

A sanity check that ties to everything else in the notes: each *column* is a vector of rotated axis directions. The image of  $(1, 0)$  is its first column  $(\cos \theta, \sin \theta)$ , and the image of  $(0, 1)$  is the second column  $(-\sin \theta, \cos \theta)$ . Both have length 1 and are orthogonal (“orthonormal columns”), and

$$R(\theta)^T R(\theta) = I, \quad \det R(\theta) = 1.$$

## 6 Two things worth naming

- **Rotation keeps the origin fixed.** We rotated about  $(0, 0)$ ; the translation freedom (“position”) is not part of  $R$ . Adding a translation  $t$  gives the full rigid motion  $p' = Rp + t$  — matching the free/fixed and pose discussion:  $R$  is the *orientation* part,  $t$  the *position*.
- **$\theta$  is the same knob we met as the *angle* in the dot product.**  $\|u\|\|v\|\cos \theta$  and this  $R(\theta)$  share the same  $\theta$ . Rotation is about preserving length and angle — which is exactly why the  $L^2$  norm and the dot product are the “rotation-friendly” tools.